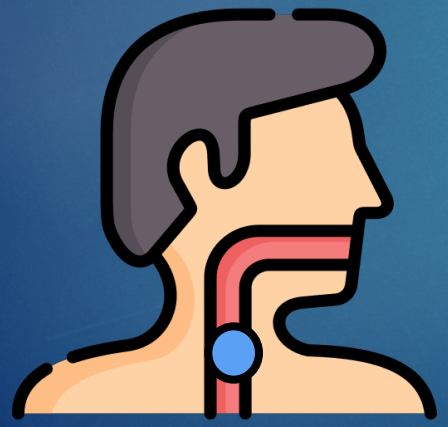
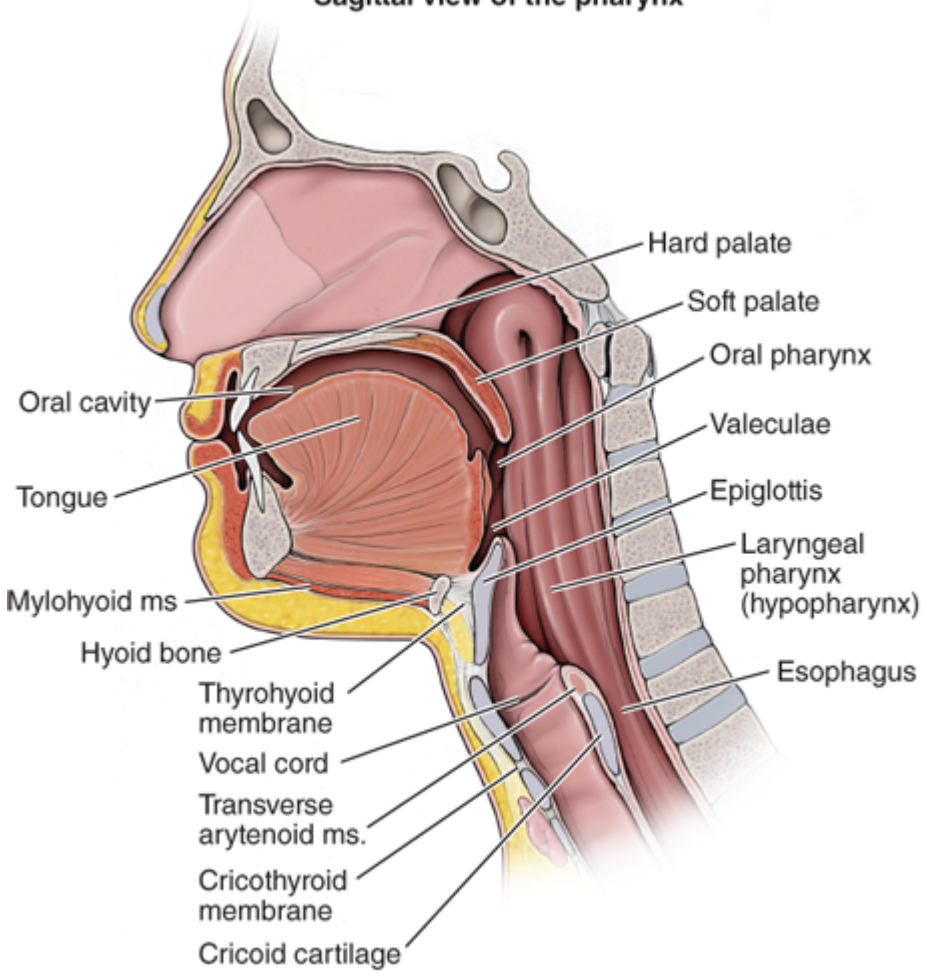
Swallowing & Dysphagia



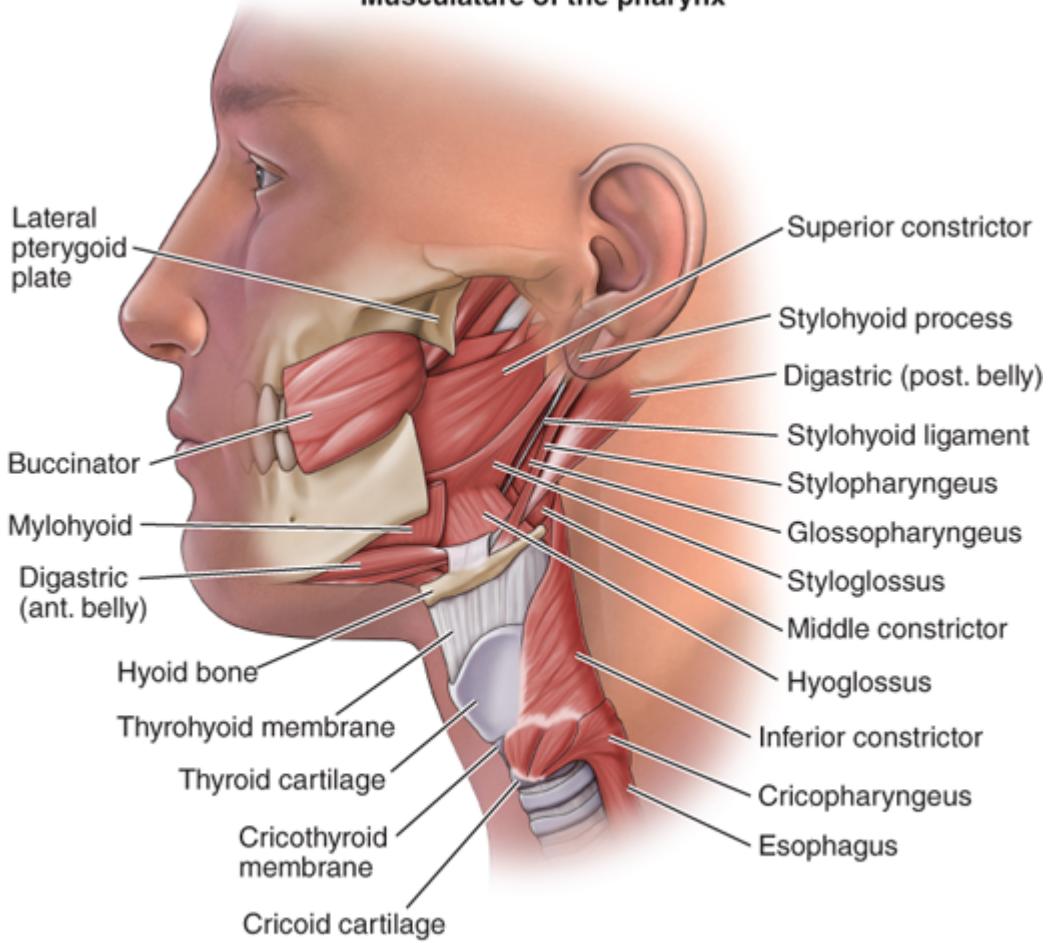
PAKKAPON RATTANACHAISIT, MD

Physiology of swallowing

Sagittal view of the pharynx

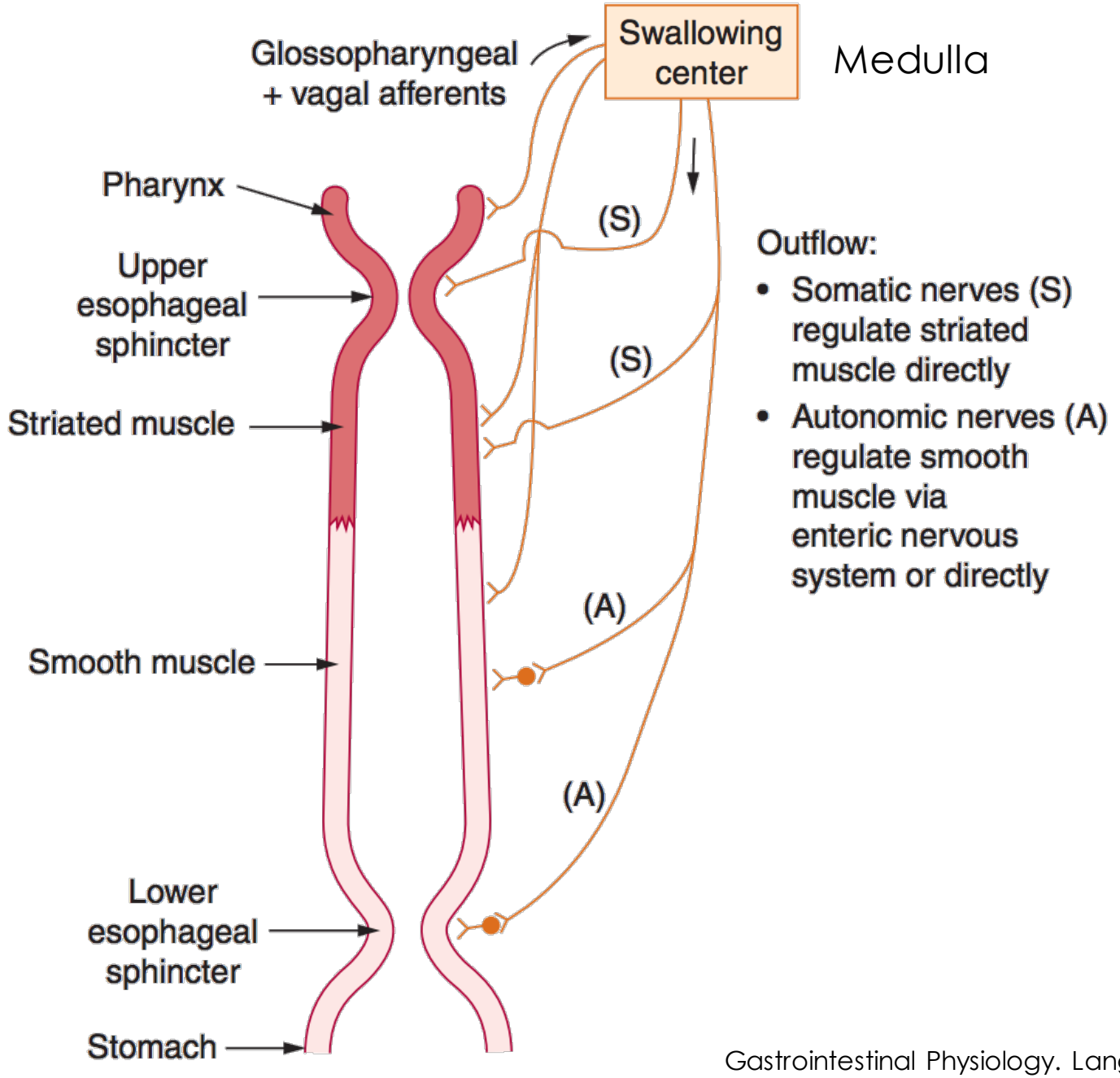


Musculature of the pharynx



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
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Functional anatomy and innervation of the esophagus

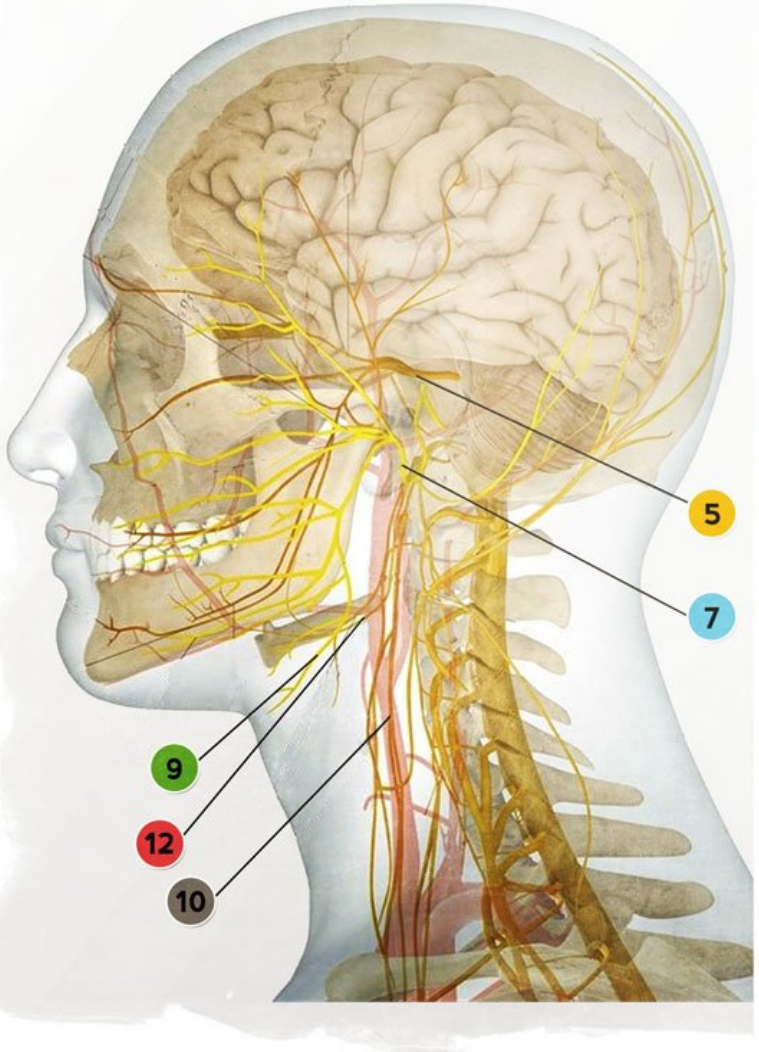


Gastrointestinal Physiology. Lange Medical Books/McGraw-Hill, 2006

THE CRANIAL NERVES AND THE SWALLOW

Cranial nerves are the nerves that emerge directly from the brain. This is in contrast to spinal nerves, which emerge from segments of the spinal cord. Generally, the left side of the brain controls the right side of the body. The right side of the brain controls the left side of the body.

However, when it comes to swallowing, many of these muscles receive bilateral innervation, meaning the brain has a much higher chance of recovering swallow abilities after unilateral brain damage, such as a stroke, or an injury to one side of the head.



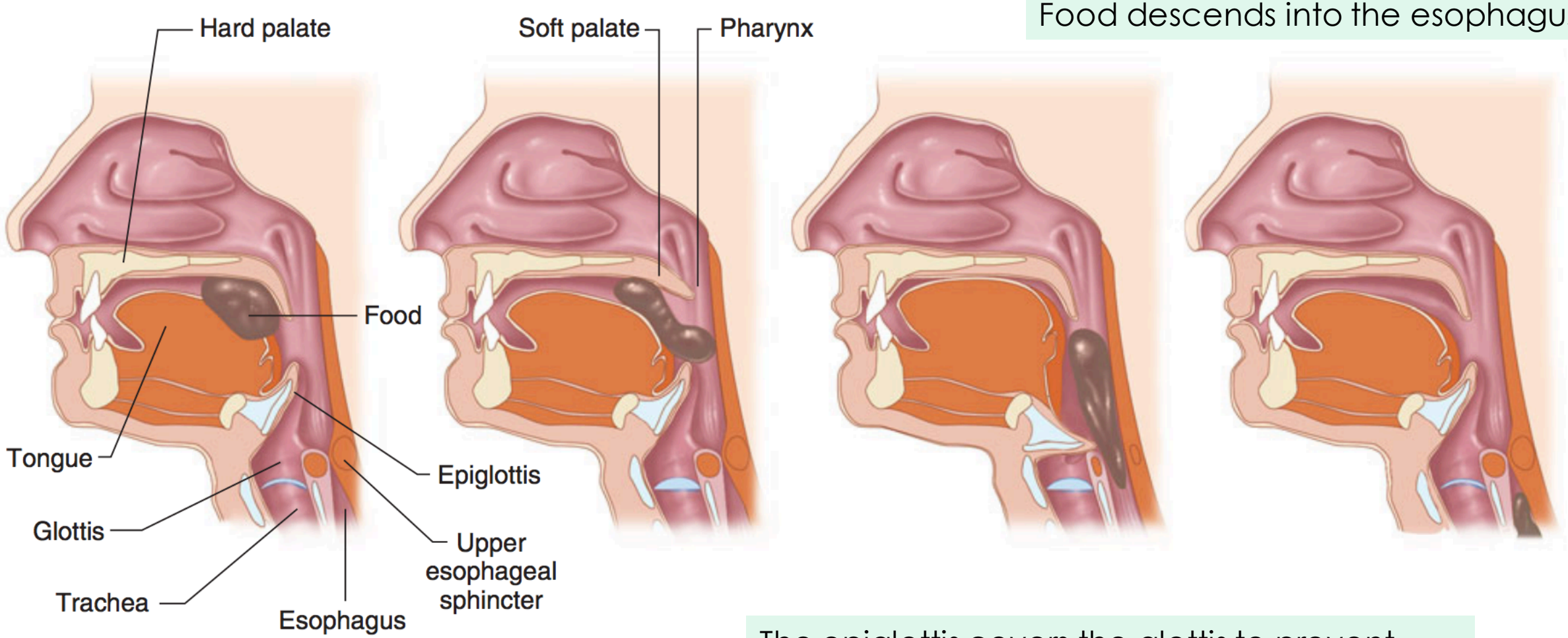
CRANIAL NERVE		IF DAMAGED
5	**TRIGEMINAL NERVE** Controls muscles of biting and chewing; provides sensation to face, cheeks, lips, jaw, forehead, eyes, eyebrows, nose, including pain, temperature, touch, and proprioception. Assists with upward/anterior movement of larynx, backward movement of tongue to soft palate, tensing the soft palate, and posterior pharyngeal wall constriction.	Loss of sensation; inability to move mandible (lower jaw).
7	**FACIAL NERVE** Provides the sense of taste in the front two-thirds of the tongue; controls the muscles of the face, including the circular muscles around the lips that assist with lip closure and keeping food/liquids in the mouth.	Paralysis of facial muscles. Poor lip strength. Dry mouth. Diminished jaw opening/closing. No taste in front 2/3 of tongue.
9	**GLOSSOPHARYNGEAL NERVE** Transmits sensation to the tongue, pharynx, and soft palate (soft part of roof of mouth). Provides sense of taste to back 1/3 of tongue. Related to dry mouth in response to fear and salivation in response to smelling food.	Decreased sense of taste and salivation; diminished or inhibits gag reflex. Weak cough reflex.
10	**VAGUS NERVE** Controls sensation of larynx, base/back of tongue, pharynx, palate, and their muscles.	Difficulty swallowing, nasal regurgitation, reduced or lost gag reflex. Hoarse, breathy, wet, voice. Inability to vary pitch.
12	**HYPOGLOSSAL NERVE** Controls the muscles of the tongue. If damage is to the nerve itself (a lower motor neuron lesion), the tongue will curve toward the damaged side. If the damage is to the nerve pathway (an upper motor neuron lesion) the tongue will curve away from the side of damage, and will occur without fasciculations or wasting, with speech difficulties more evident.	Inability to position food for chewing, resulting in food getting pocketed in cheeks.

Chewing

- Decreases the size of food particles to facilitate swallowing and to begin the digestive process
- Lubricates food by mixing it with saliva

The soft palate elevates to prevent food from entering the nasal passages.

Food descends into the esophagus.



The tongue pushes the food bolus to the back of the mouth.

The epiglottis covers the glottis to prevent food from entering the trachea, and the upper esophageal sphincter relaxes.

Vander's Human Physiology, 11th ed. McGraw-Hill, 200

Stage of swallowing

1. Oral (Voluntary) phase: Bolus moves from oral cavity into the oropharynx
   - lip closure, inhibit respiration, tongue pushes the bolus to the back

2. Pharyngeal phase: Bolus moves from the oropharynx into the esophagus; involuntary process
   - Soft palate lifts (close nasal airways), epiglottis is pushed backward, larynx and vocal folds contract, UES open

3. Esophageal phase: Bolus moves through the esophagus and into the stomach; involuntary process
   - Food bolus is propelled down the esophagus by peristalsis


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Dysphagia

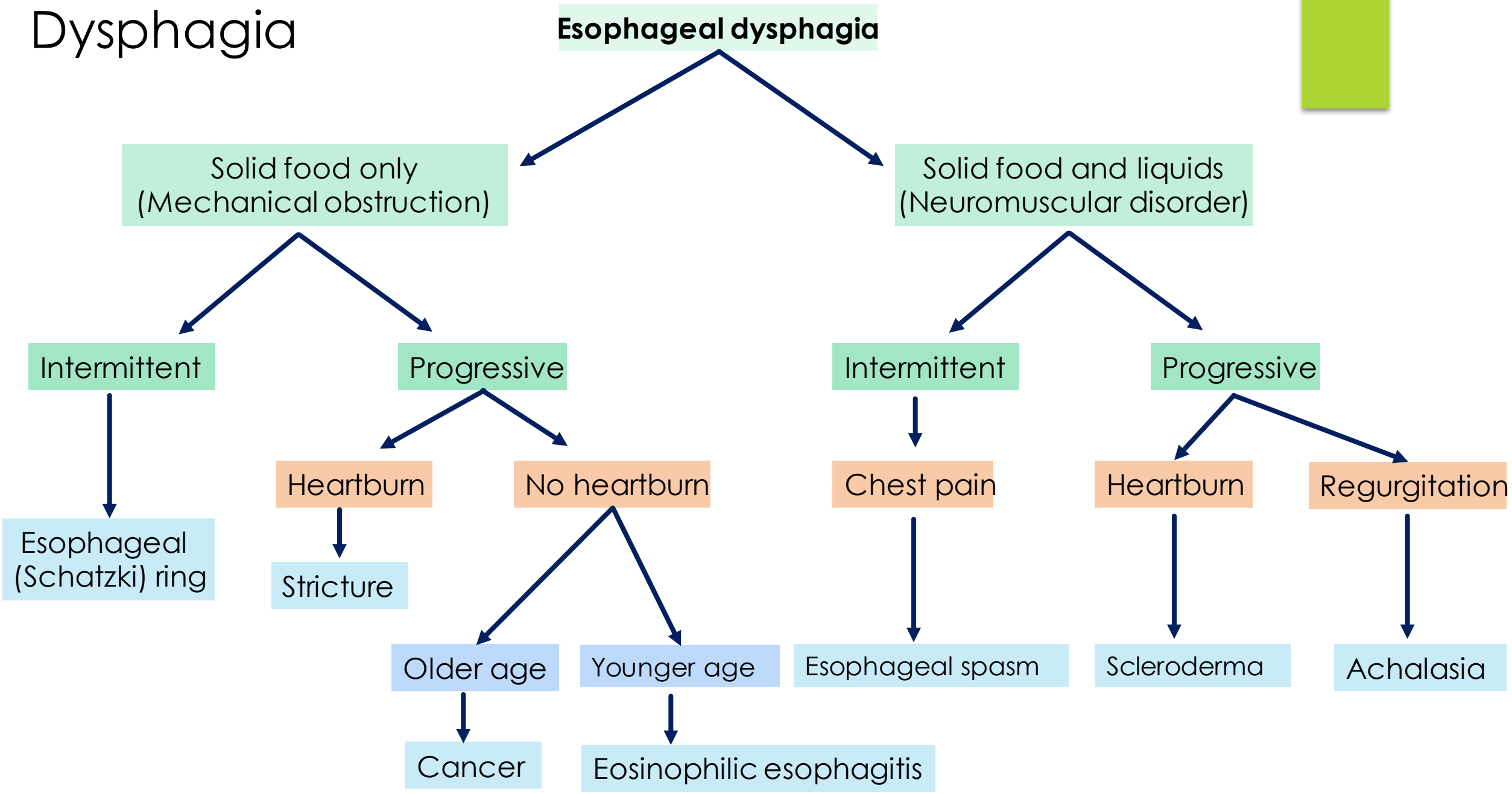
Oropharyngeal dysphagia

- Abnormalities affecting the neuromuscular mechanism of the striated muscle of the mouth, pharynx, UES
- Dysphagia localized to neck, nasal, regurgitation, aspiration, assoc. ENT symptoms
- Can't initiate swallow

Esophageal dysphagia

- Caused by the variety of disorders affecting the smooth muscle of esophagus
- Dysphagia localized to chest or neck, food impaction
- Swallowed, but food is stuck

Dysphagia


Esophageal dysphagia
Solid food only (Mechanical obstruction)
Solid food and liquids (Neuromuscular disorder)
Intermittent
Progressive
Intermittent
Progressive
Esophageal (Schatzki) ring
Heartburn
No heartburn
Chest pain
Heartburn
Regurgitation
Stricture
Older age
Younger age
Esophageal spasm
Scleroderma
Achalasia
Cancer
Eosinophilic esophagitis